

School choice and parents' preferences for school attributes in Chile

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ABSTRACT

A key assumption of school choice and competition policies is that parents' most important (if not only) priority when choosing a school is its quality. However, evidence about which of a school's attributes really drives parental choice is still scarce. We use census data from a parent questionnaire in Chile, a country with a national school choice and competition system, to describe the attributes most commonly considered by parents when choosing a school, and to assess how the probability of prioritizing those attributes varies with the parents' socioeconomic characteristics, while controlling for other characteristics of the family. We find that parents choosing a school prioritize its proximity, its quality, and whether it provides religious education. Furthermore, the probability of parents prioritizing proximity is higher for parents of low socioeconomic status, while the probability of them prioritizing quality and religious education is higher for parents of high socioeconomic status. These findings show that only advantaged families choose schools based on their quality, and therefore school choice and competition policies may offer a limited benefit for disadvantaged pupils, possibly maintaining or reinforcing socioeconomic segregation in the education system.

1. Introduction

A long-standing debate in education policy is whether free choice and competition benefit educational systems. On the one hand, advocates of school choice programs argue that in a competitive educational system, parents choose the highest-quality schools, considering quality as an objective indicator (standardized test results). According to this perspective, the educational system can only improve its efficiency and fairness by increasing “free choice” among families of different socioeconomic status (Betebenner, Howe, & Foster, 2005; Robertson, 2000). Competition between public and private-voucher schools would increase incentives to satisfy parents' expectations, reducing drop-out levels, increasing graduation rates and improving school quality. In this scenario, low-performance schools would eventually disappear, increasing the number of high-quality schools available, especially for low-income families that cannot afford private schools. Therefore, the system would promote efficacy and innovation in the entire educational system (Friedman, 1962; Levin, 2002).

On the other hand, opponents of free school choice programs present evidence that families with low socioeconomic status do not consider standardized test results when choosing a school. They usually take

into account non-academic variables, such as proximity or composition, and this situation may promote socioeconomic and academic segregation in the long term (Kremer & Sarychev, 2000; Ladd, 2002; Levin, 1998; Schneider, Buckley, & Elacqua, 2006). Moreover, public and private-voucher (subsidized) schools face different circumstances in a free school choice system. While public schools cannot select which pupils they take, private-voucher schools can use academic tests and admission requirements, for instance, to improve the average ability level of their pupil body, thus promoting a “brain drain” from public schools (Hsieh & Urquiola, 2003; Levin, 1998; McEwan, 2000; Neal, 2011). Furthermore, private-voucher schools may be perceived to be high-quality schools, attracting families with higher social and cultural capital and encouraging their exit from the public system (Hirschman, 1970). Considering that families with greater resources are those who potentially have a greater ability to foster improvements in the school system, their exit is a serious loss for the public sector (Campbell, West, & Peterson, 2005).

A central aspect of this debate is whether parents choose schools based on their quality. Is quality the most important factor when parents choose a school? How do parents' preferences vary by socioeconomic level and ethnic background? The evidence collected on this

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topic is not conclusive because school choice programs have been implemented in unique contexts with very different geographic and socioeconomic characteristics. Moreover, in the cases where data is available, researchers have focused on comparing performance between public and voucher/charter schools (Berends & Waddington, 2016; Torche, 2005); less attention has been devoted to understanding the decision-making process behind parents' choices, which is a central aspect in explaining how school choice programs work. Furthermore, most of the evidence available on parents' preferences refers to developed countries, such as the United States and England, or to targeted school choice programs; the conclusions therefore cannot be generalized either for developing countries or for universal school choice programs.

This article contributes to the literature on school choice and competition, providing evidence about parents' preferences for school attributes in a school choice program implemented in a developing country, namely Chile. In Chile, families can choose between public and private-voucher schools, regardless of their socioeconomic background. Additionally, the Chilean voucher system has operated nationwide, almost unaltered, for more than 35 years, and Chile's extensive database offers an ideal source for analysis of the free school choice program (Carnoy & McEwan, 2000).

This article analyzes parents' preferences for school attributes using census data (nationwide) in Chile, and explores what factors are considered by families of different income levels, educational attainment and ethnic background when they choose a school. The article has five sections. The following section introduces the Chilean case and reviews research conducted in the US, Europe, and Chile. The third section discusses the data and methods used. The fourth section presents the results and the final section contains the discussion and conclusions.

2. Literature review

What do parents consider when they are choosing a school for their children? Parents take into account several aspects, such as school quality, religious or moral instruction (Trivitt & Wolf, 2011), and whether the school is suitable for their children (Teske & Schneider, 2001) or safe (Kelly & Scafidi, 2013). Even where quality is the principal aspect that parents consider when choosing a school (Hamilton & Guin, 2005), there is no single definition of quality. While some parents define it based on teacher quality (DiPerna & Catt, 2016a; Stewart et al., 2010; Wasley et al., 2000), others assume school quality is reflected in test scores (Schneider, Teske, & Marschall, 2002), or other academic aspects (Catt & Rhinesmith, 2016; Lincove, Cowen, & Imbrogno, 2018). Parents may also define quality based on school characteristics such as class size (Kelly & Scafidi, 2013; Stewart, Wolf, Cornman, McKenzie-Thompson, & Butcher, 2009), extracurricular activities (DiPerna & Catt, 2016b; Kelly & Scafidi, 2013; Lincove et al., 2018), and school composition – the socioeconomic, cultural and ethnic characteristics of the pupil body (Dee & Fu, 2004; Ladd, 2002) –. Last but not least, the definition of school quality might be based on the information parents gather from informal mechanisms such as comments or suggestions from friends, family or neighbors (Ball & Vincent, 1998).

Furthermore, current evidence shows that parents weight these factors differently depending on their socioeconomic or racial background. Qualitative studies in England conclude that race is an important aspect for white parents who wish to avoid multi-ethnic environments (Bagley, 1996). Similar studies in Alberta and England show that parents' definition of school quality differs with family social networks and income level (Bosetti, 2004), as well as cultural background (Ball et al., 1995). Moreover, while high-income families choose

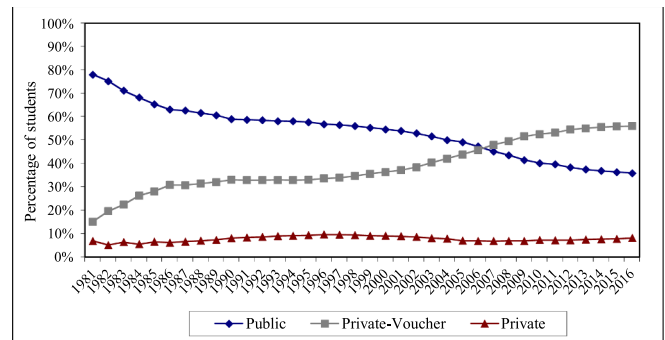


Fig. 1. Enrollment in primary and secondary education in the Chilean system (1981–2016).

schools based on academic criteria, low-income families prefer nearby schools (Ball, Bowe, & Gewirtz, 1996). A common challenge for all these studies is to establish, at the aggregated level, whether parents of different socioeconomic status have diverse preferences for school attributes. In fact, most studies are qualitative or analyze school choice programs which are only available to low-income families (Erickson, 2017).

A remaining question is whether the free school choice system is equally beneficial for all families. Studies suggest that wealthier families may be more likely to take advantage of this system because they have better access to (high quality) information (Broccolichi & Zanten, 2000; Van Zanten, 2003), greater ability to afford transportation, and more experience with choices and alternatives (Levin, 1998). In this respect, evidence from England demonstrates that middle-class parents take “full advantage” of the educational market (Ball et al., 1995). Low-income families, on the other hand, are at a disadvantage because they have less access to information, are less likely to use data from standardized tests to evaluate the quality of schools, and are more likely to prioritize non-academic aspects in selecting a school (Ladd, 2002; Schneider et al., 2006).

This study fills a gap in the literature by analyzing parents' preferences in selecting a school in a universal and well-established free school choice program in a developing country, using representative data at the national level. Furthermore, it compares those preferences in parents from different socioeconomic and ethnic backgrounds, and analyzes whether their preferences vary for different levels of their children's education.

2.1 Background: The Chilean educational system

Under the military regime of Augusto Pinochet (1973–1990), Chile implemented one of the most radical and ambitious neoliberal reforms in Latin America. These reforms included the introduction of new labor legislation, the transformation of social security, the privatization of health care, the internationalization of agriculture, the transformation of the judicial system, the decentralization and regionalization of government administration and the privatization of education (Silva, 1991).

The educational reform introduced in Chile in 1981 included two main aspects: first, it shifted the administration of education from the central government to local governments (municipalities); and second, it shifted the public (free) educational system towards the private sector, introducing a new actor, private-voucher schools. Supporters of the reform claimed that this privatization would eliminate the state monopoly, and that competition would, eventually, improve the quality of the entire educational system (Gauri, 1998).

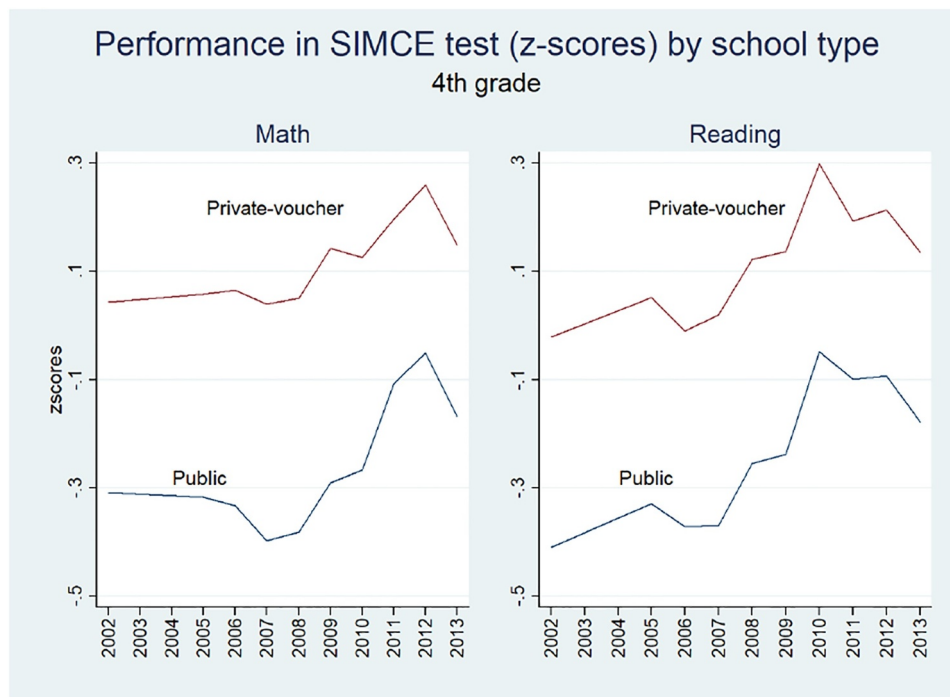


Fig. 2. Performance in SIMCE test by school type in 4th grades (Standardized test scores).

In this model, parents who opt for the public education system can choose between a municipal or a private-voucher school¹. Public funding is defined by pupils' attendance; therefore the revenue of each school is determined on a monthly basis (Carnoy & McEwan, 2000). As "funds follow the pupil" (Mizala & Romaguera, 2000), schools compete to enroll as many pupils as they can (Torche, 2005). This reform radically changed the distribution of pupils by type of school. For example, in 1981, when the system was implemented, 78 percent of pupils from elementary, middle and high school attended public schools. However, in 2016, due to the sustained growth of voucher school enrollment (from 15 to 56%), only 36 percent of pupils remained in the public system (Fig. 1).

The reasons behind parents' preference for private-voucher schools are not yet clear. Advocates of free school choice argue that parents base their decision on school quality measured through the schools' performance in standardized tests. Evidence for this argument might be reflected in Fig. 2, which shows that the averages in math and reading in the national standardized test² (SIMCE) are higher for private-voucher schools (almost 0.15 standard deviations above the national average). However, previous studies have suggested that parents choose schools based on other criteria, such as socioeconomic composition, religious education and proximity (Raczynski et al., 2011; Treviño et al., 2016). The following section discusses the evidence for this.

2.2 Parents' preferences for school attributes in Chile

The available evidence suggests four important aspects that parents consider when selecting a school. First, although most parents consider school quality, the parental definition of school quality is not based on standardized test results (Schneider et al., 2006; Taut et al., 2009), and there is no direct relation between parents' preferences and school effectiveness (Carrasco and San Martín, 2012). Second, one of the main aspects that parents consider when choosing a school is geographical proximity, not only in Santiago (Chumacero, Gómez, and Paredes, 2011) but also in other regions such as Coquimbo, Araucanía and Los Lagos (Gubbins, 2013; Hernández & Raczynski, 2015). Third, studies in the Santiago and Valparaíso Regions show that parents also consider associated costs when choosing a school (Román, 2014; Gallego and Hernando, 2009), and the school's composition. Families in Santiago prefer a pupil body from their own social class or a higher one, but for different reasons. While low-income families seek to avoid schools with serious behavior problems (Córdoba, 2014), middle-class families choose private-voucher schools to prevent their children from associating with low-income pupils, which they consider risky (Canales, Bellei, & Orellana, 2016), and upper-middle-class families choose specific schools in order to remain in or become part of the elite (Kosunen & Carrasco, 2016).

Furthermore, parents' preferences seem to vary by socioeconomic level. Alves et al. (2015) found that as family income increases, the probability of choosing a school outside of their neighborhood also increases. There is also an interaction between parents' preferences and the admission barriers imposed by private-voucher schools, such as admission tests or additional fees³, which increases school segregation in Santiago (Santos and Elacqua, 2016). There, upper-income families are surrounded by high-quality schools with high tuition fees, while low

¹ Private-voucher schools can apply specific admission policies (including academic tests, religious commitments, etc.), and can charge tuition and monthly fees. In 2015, Bachelet's government approved the Inclusion Law (Law number 20845), which eliminates selection criteria, co-payment and profit for private-voucher schools. The law has been implemented gradually since 2016 and therefore is not yet applied nationwide.

² In Chile these tests are known as SIMCE, which stands for *Sistema Nacional de Evaluación de Calidad de la Educación* (National Educational Quality Measurement System). In this article we use standardized test or SIMCE interchangeably.

³ Since 1993 private-voucher schools have been allowed to charge additional tuition over and above the voucher that they receive from the government. This payment constrains the pool of schools that parents can choose, especially among low-income families. This co-payment is being gradually eliminated by the Inclusion Law (2015) mentioned above.

income families are restricted to nearby, affordable, lower quality schools. In this context, the availability of schools (Flores & Carrasco, 2013) as well as private-voucher schools' admission requirements (Román & Corvalán, 2016) largely restrict low-income parents' preferences.

These studies conducted in Chile have either used administrative data collected by the Ministry of Education or information collected through parent interviews or surveys. Despite the different sources and methods used, the majority of these studies are focused on parents' school choices in primary education, in the metropolitan region of Santiago (the capital of Chile), which concentrates 38% of pupils in primary and secondary education (Mineduc, 2016), and in specific socioeconomic groups (middle or low-income families).

The school choice program may work very differently in Santiago, because, on average, families have higher levels of income, education, connectivity and access to basic services, than in the rest of the country (Ministerio de Desarrollo Social, 2016).

This study overcomes these limitations by: incorporating national data from all municipalities in Chile, for four different school grades: 2nd, 4th, 8th and 10th grades⁴, in all schools in the country. We aim to answer the following research questions: a) What are the main school attributes that parents consider when they choose a school?; b) Do the preferred attributes vary by family income, parents' education and ethnicity?; and c) Do parents' preferences change as pupils progress from elementary to middle school and high school? To answer these questions, we use the following data.

3. Data and methods

We used data from the *Sistema Nacional de Evaluación de Calidad de la Educación* (National Educational Quality Measurement System), also known as SIMCE. These standardized tests measure pupil achievement in reading, math, sciences, and social sciences in 2nd, 4th, 8th and 10th grade. SIMCE is administered in every Chilean school, and since 2005 it has been administered at the end of every school year (which runs from March through December). The test includes a self-administered parent questionnaire which measures families' demographic and socioeconomic characteristics (Contreras, Bustos, & Sepulveda, 2010). The questionnaire has an 85% response rate (Mineduc, 2013).

For this study, we considered five variables from the parent questionnaire: parents' preferences, i.e. factors or reasons that parents consider when choosing a school; father's education; mother's education; family income and ethnicity. The descriptive statistics of each of these variables are presented in Table 1

- (a) Parental preferences: Our main dependent variable is based on the question "What is the main reason why you enrolled your child in this school?" Parents can choose their answer(s) from among the following options: (1) school proximity, (2) school infrastructure, (3) pupil has friends in the school, (4) religious education, (5) school quality, (6) school has similar pupils to my child, (7) it is a low cost school, (8) it is the only school available in this municipality, (9) school doesn't accept pupils from other schools, (10) pupil has siblings in the school, (11) bilingual education, (12) discipline, (13) technical education and (14) other reasons. Table 2 indicates that, regardless of school grade, the following are the three most important reasons that parents consider when choosing a school: school proximity, school quality and religious education.
- (b) Parents' educational attainment: In the question "What is the

⁴ In Chile, Primary school includes 8 years of schooling (1st to 8th grade), and Secondary School is composed of 4 years (9th to 12th grade). In this study, parental school preferences are analysed for the equivalents of elementary school (2nd and 4th grade), middle school (8th grade) and high school (10th grade).

Table 1

Parents' characteristics in 2nd, 4th, 8th and 10th grades (2012).

		2nd	4th	8th	10th
Years of schooling (mean)	Father	11.9	11.6	11.3	11.7
	Mother	11.8	11.6	11.2	11.6
Parents' ethnicity	Father indigenous	12%	15%	13%	10%
	Mother indigenous	15%	17%	16%	12%
Family income (monthly) ^a	< US \$223	9%	9%	10%	7%
	US \$224 - 446	26%	27%	30%	24%
	US \$447 - 670	20%	20%	20%	20%
	US \$671 - 893	11%	11%	11%	12%
	US \$894 - 1,116	7%	7%	7%	8%
	US \$1,117 - 1,340	5%	5%	5%	6%
	US \$1,341 - 1,786	5%	5%	4%	5%
	US \$1,787 - 2,233	4%	3%	3%	4%
	US \$2,234 - 2,680	2%	2%	2%	2%
	US \$2,681 - 3,127	2%	2%	1%	2%
	US \$3,128 - 3,573	1%	1%	1%	1%
	US \$3,574 - 4,020	1%	1%	1%	1%
	> US \$4,021	7%	6%	5%	7%

^a For easier interpretation family income has been converted from Chilean pesos (2012) to US dollars (2017), according to the American Bureau of Labor Statistics consumer price index.

Table 2

Main parents' preferences for school attributes in 2nd, 4th, 8th and 10th grades (2012).

	2nd	4th	8th	10th
School proximity	36%	38%	43%	23%
School infrastructure	3%	3%	3%	3%
Student has friends in the school	2%	1%	2%	3%
Religious school	15%	14%	12%	14%
School quality	19%	17%	14%	20%
School has similar students that my child	4%	4%	4%	3%
Low cost	2%	2%	3%	3%
It is the only school in the municipality	1%	1%	1%	1%
School doesn't accept students from other schools	1%	1%	1%	2%
Student has siblings in the school	7%	6%	5%	4%
Bilingual education	1%	1%	1%	1%
Discipline	5%	6%	5%	6%
Technical education	1%	1%	1%	11%
Other reasons	5%	5%	5%	5%
Total (N)	137,004	191,606	188,331	152,955

- highest educational level attained by the pupil's mother and father?", parents select answers ranging from 1st grade elementary school to a Ph.D. degree. We transformed these 20 categories into four groups: elementary, middle, high school and college. Also we created the variable years of schooling, using the guideline created by the Ministry of Education⁵ (Mineduc, 2013). In our sample, parents have completed an average of 11 years of schooling.
- (c) Family income: In the question "In a normal month, what is the total income of your household?", family income was classified into 13 categories, where the lowest household income is lower than CH\$100,000 (USD\$223 in 2017) and the highest household income is higher than CH\$1,800,001 (USD\$4,021 in 2017). Table 1 shows that across different grades, most families (more than 60 %) earn a monthly income below USD\$893. Using the method developed by Von Hippel, Hunder and Drown (2017) we transformed "family income" from a categorical variable to a quantitative variable and estimate income quartiles, based on the latter. We do

⁵ Check Table 1 in the report "Metodología para la construcción de grupos socioeconómicos. Pruebas SIMCE 2013" http://archivos.agenciaeducacion.cl/Metodologia_de_Construccion_de_Grupos_Socioeconomicos_Simce_2013.pdf

⁶ Chilean Pesos.

this transformation converting family income to the midpoint of each category, with the exception of families at the top income category. For this category, we used the harmonic mean of a Pareto distribution. Using the midpoints and the harmonic mean of Pareto distribution allows us to estimate accurate income statistics. This method estimates mean incomes that are 99 percent reliable with less than 2 percent of bias (von Hippel, Hunter, & Drown, 2017).

- (d) Ethnicity: Parents answered “Yes/ No” to the question “Do the pupil's parents (father and mother) belong to any of the following indigenous groups?”. Table 1 shows that the percentage of parents that identify themselves as “indigenous” is around 13 percent.

3.1 Methods

We approach our analysis in two stages. First, we use descriptive statistics to explore the relationship between parents' preferences, family income, parents' education, and ethnicity. Second, we ran three different logistic regression models to estimate the influence of family income, parents' education, and ethnicity on the three most frequent attributes that parents consider when choosing a school: “school proximity”, “school quality” and “religious education”.

$$Pref_p = \alpha_p + QInc_p + Educ_p + Indigenous_p + Outside_p + \omega_p + e_p$$

The dependent variable is the parents' preference for school attributes ($Pref_p$) with value 1 if parents consider a specific school attribute when they choose a school (proximity / religion / school quality) and 0 otherwise. Families' socio-demographic characteristics are included in the model with three variables: (i) income quartiles ($QInc_p$), which is a categorical variable where the lowest quartile (quartile 1) is the baseline; (ii) parents' educational level ($Educ_p$), we used as a baseline those parents with elementary school; and (iii) parents' ethnicity, classified in three dummy variables ($findigenous$ if the father is indigenous, $mindigenous$ if the mother is indigenous, and $fmindigenous$ if both parents are indigenous). Finally, we included a measure of whether the pupil attends a school outside the municipality where he/she lives ($Outside_p$). This is a proxy for schools' proximity to pupils' households.

The rest of control variables are represented by ω_p . Which includes three school characteristics: whether the school is located in an urban or rural area ($Urban_p$); whether the school is public, private-voucher or non-subsidized private (Sch_p), considering public schools as the baseline; and we used a dummy variable for each municipality. In consequence, parents' preferences were compared to each other within the same municipality (fixed effects at municipality level).

Furthermore, the model includes three variables to control for aspects that could limit parents' options to choose a school. First, $Selection_p$ identifies whether pupils were screened before being enrolled (through interviews, reports of parents' marital status or income, academic tests, etc.). This dummy variable is represented by 1 if more than 50 percent of parents report that they went through a selection process in their school, and 0 otherwise. Second, we include the costs of each school, adding in the model the dummy variable $Fees_p$ with value equal to 1 if the school charges additional fees and 0 otherwise. Third, we include the total number of schools available in each municipality. Finally, in order to control for the local school supply, we only include municipalities with at least three schools available for each educational level. This reduces the sample from 345 to 328 municipalities in 2nd grade and 4th grade models, 311 municipalities for 8th grade models, and 180 municipalities for the analysis with 10th grade pupils.

4. Results

4.1. Descriptive analysis

As we mentioned earlier, the three most important aspects that parents consider when choosing a school are proximity, quality and

religious education. These attributes explain approximately 70% of parents' preferences (Table 2), with “proximity” being the most important attribute; its importance increases from 36% in elementary school to 43% in middle school, but declines to 23% in high school. The second most important attribute is quality, the importance of which decreases from 19% in elementary to 14% in middle school, and increases again in high school (20%). Finally, the third most reported attribute is religious education, which remains around 14% among elementary, middle and high school pupils. The remainder of the school attributes only explain about one-third of parental preferences.

As Figs. 3–7 show, proximity is the most important school attribute for low-income families, parents with low education levels and indigenous parents. Furthermore, those figures show that as income and years of education increase, the importance of school proximity decreases. For instance, Fig. 3 shows that the importance of school proximity in second grade decreases from more than 50% in the lowest income category to about 10% in the highest income category.

Fig. 3 also shows that this trend is similar in all grades. Figs. 4 and 5 show very similar trends for father's and mother's years of schooling, respectively. For parents with the fewest years of education, proximity has a relative frequency of about 60% among the reasons for choosing a school. That percentage decreases to about 20% for parents with the most years of education. Figs. 6 and 7 show that indigenous parents tend to value school proximity more highly than non-indigenous parents, although in this case the difference is smaller, around 5% in 2nd, 4th and 8th grades; in 10th grade there does not appear to be any difference.

As family income and parents' educational level increase, school quality and religious education become more important. For instance, when comparing across income levels, the importance of religious education and school quality increases from about 5% for the lowest income level to about 15% for the highest income level among pupils in all grades. Figs. 4 and 5 show a very similar trend when comparing parents' years of education. In summary, school quality and religious education are more important factors when choosing a school than school proximity for parents with higher income levels and more years of education.

In the case of indigenous parents, school quality and religious education are less important attributes than for non-indigenous parents. Other attributes seem to play a bigger role in the decision process of indigenous parents (Figs. 6 and 7).

4.2. Logistic regression models

We ran logistic regression models to estimate the influence of family income, parental education and ethnicity in each of the three most important school attributes, for 2nd, 4th, 8th and 10th grades, while controlling for demographic and school characteristics that influence the relationship between socioeconomic characteristics and parents' preferences. In total, we ran 12 models – three logistic regression models for each of the four school grades that we include in our analysis.

Model 1 (school proximity) shows that low-income families (quartile 1) prioritize school proximity, especially in elementary school. However, as family income increases, proximity becomes less important as pupils progress through the system (Table 3)⁷.

High income families (quartile 4) are 28%, 30%, 22% and 16% less likely to choose a school based on its proximity in 2nd, 4th, 8th and 10th grade, respectively ($p < 0.05$), compared to low income families (quartile 1). Similarly, as parents with college education are less likely to choose a school based on its proximity, and this trend is very similar in elementary, middle and high school levels ($p < 0.05$). Finally, children whose both parents are indigenous are more likely to attend a

⁷ Table 5 uses as baseline income quartile 2, please see the appendices.

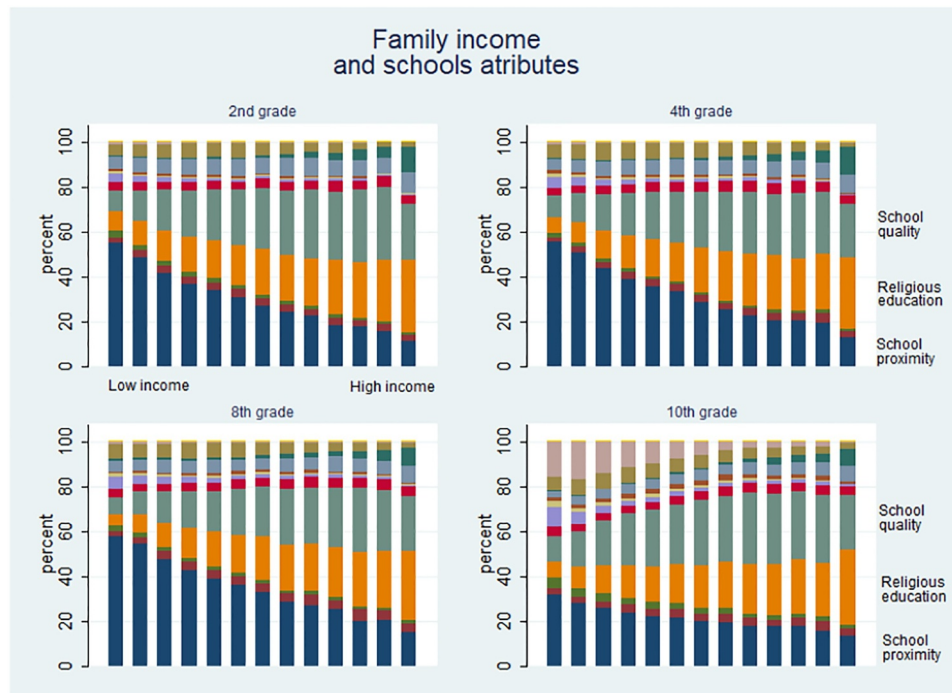


Fig. 3. Relationship between family income and preferences for school attributes.

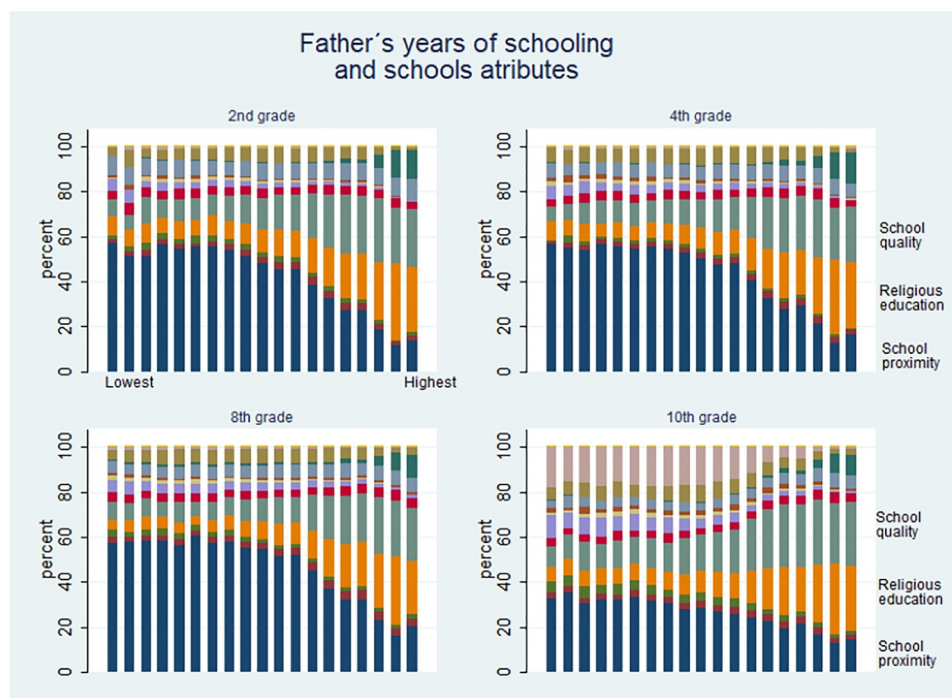


Fig. 4. Relationship between father's education and preferences for school attributes.

school close to home in elementary school ($p < 0.05$).

Model 2 (school quality) shows that as family income increases, the likelihood of choosing a school based on its quality also increases, but only since 4th grade (Table 3). With the exception of 2nd grade, in all grades families in higher income quartiles are more likely to choose a school based on its perceived quality as compared to low income families (quartile 1). High income families (quartile 4) are 47% more likely to choose a school based on its quality in 4th grade, 44% in 8th grade and 67% in 10th grade ($p < 0.05$).

As parents' education increase, the likelihood of choosing a school based on its quality also increases ($p < 0.05$) in every grade. Finally, the difference between indigenous and non-indigenous parents are not statistically significant ($p > 0.05$).

Model 3 (religious education) indicates that the likelihood of choosing a school due to its religious education is lower among high income families (quartile 4) in elementary school, as compared to the base category (quartile 1). In every grade, as the mother's educational level increases, the likelihood of choosing a school based on its religious

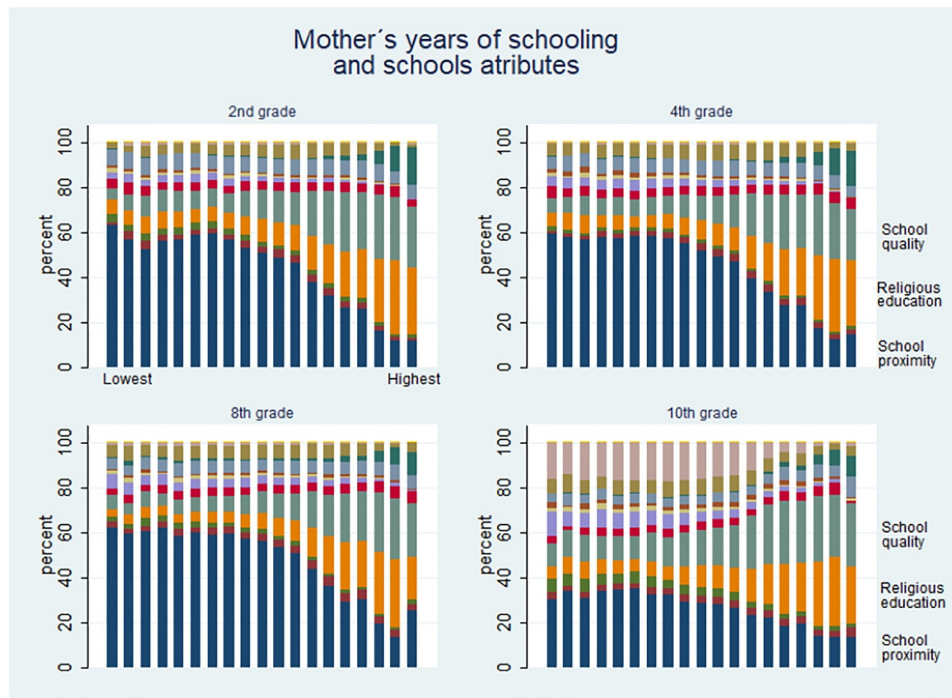


Fig. 5. Relationship between mother's education and preferences for school attributes.

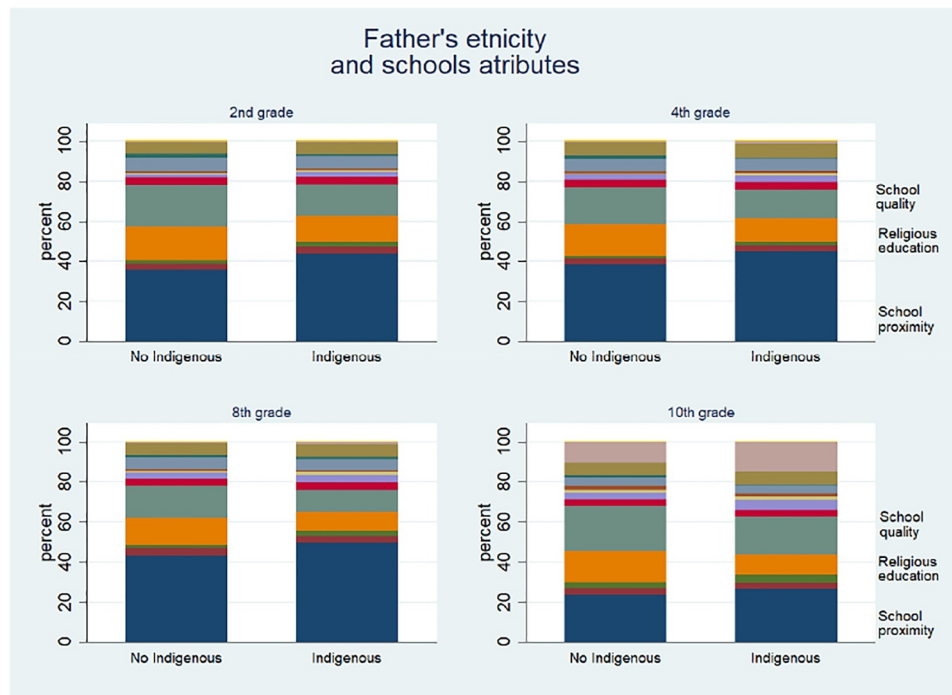


Fig. 6. Relationship between father's ethnicity and preferences for school attributes.

orientation also increases ($p < 0.05$). Finally, in 10th grade, when both parents are indigenous, they are more likely to choose a school based on its religious orientation ($p < 0.05$).

4.3. Predicted probabilities

A more intuitive way to see the relationship between socioeconomic characteristics and the prevalence of some school attributes when choosing a school is through estimation of predicted probabilities of

choosing those attributes for families with different socioeconomic characteristics, based on the results of the models discussed above. We employ two strategies for this estimation, focusing on two of the school attributes under analysis: proximity and quality.

The first strategy is to estimate average predicted probabilities of choosing a school based on proximity or quality for families across income quartiles. We perform this estimation for families in which both parents are indigenous and families in which neither parent is indigenous. Fig. 8 shows that the average predicted probability of

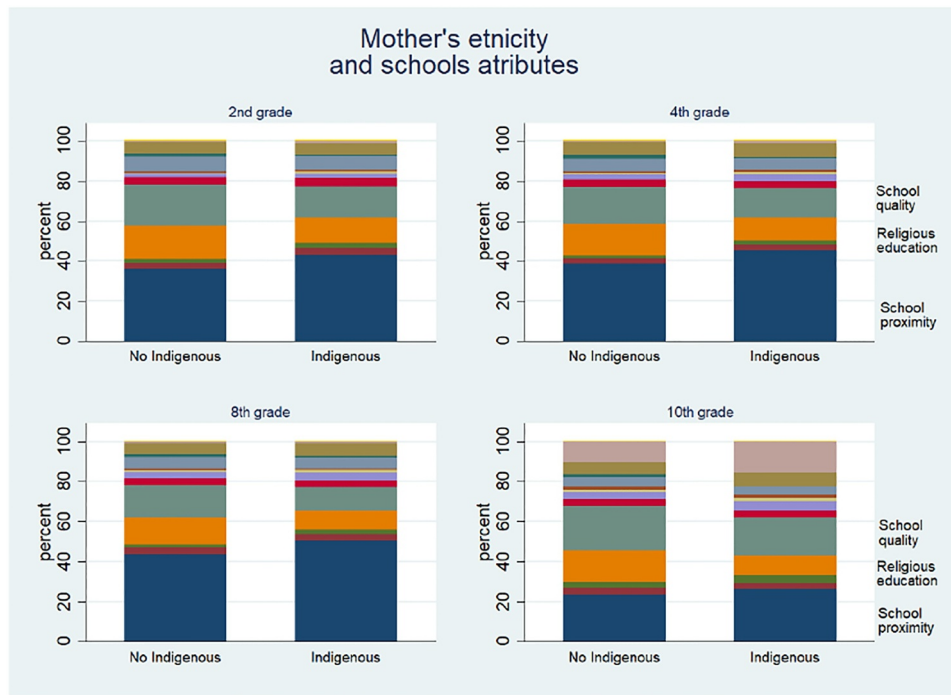


Fig. 7. The relationship between mother's ethnicity and preferences for school attributes.

Table 3

Logistic regression models (odds ratio) of parents' preferences for school attributes in 2nd, 4th, 8th and 10th grades.

	School proximity Model 1				School quality Model 2				Religious education Model 3			
	2nd	4th	8th	10th	2nd	4th	8th	10th	2nd	4th	8th	10th
Family Income												
Quartile 2	0.87***	0.87***	0.91***	0.94*	1.00	1.20***	1.16***	1.24***	1.00	1.00	1.08*	1.02
Quartile 3	0.81***	0.81***	0.84***	0.89***	0.96	1.33***	1.26***	1.47***	0.96	0.99	1.05	1.03
Quartile 4	0.72***	0.70***	0.78***	0.86***	0.88***	1.47***	1.44***	1.67***	0.88***	0.92**	0.99	0.98
Father education												
Father_middle_school	1.00	0.97	1.05 ⁺	0.93	1.29**	1.05	1.08	1.15**	0.96	1.11	1.04	1.01
Father_high_school	0.95	0.91*	1.00	0.96	1.37***	1.20***	1.23***	1.33***	0.99	1.16*	1.05	1.06
Father_college	0.86**	0.87***	0.89***	0.96	1.42***	1.29***	1.32***	1.46***	0.98	1.12	1.09	1.06
Mother education												
Mother_middle_school	0.94	0.98	0.92**	0.91*	1.08	1.13 ⁺	1.20**	1.16*	1.07	1.03	1.33***	1.11
Mother_high_school	0.73***	0.73***	0.78***	0.76***	1.59***	1.66***	1.74***	1.58***	1.28*	1.25**	1.64***	1.44***
Mother_college	0.59***	0.58***	0.60***	0.64***	1.75***	1.87***	2.01***	1.70***	1.36**	1.33***	1.77***	1.64***
Indigenous												
Father indigenous	0.95	0.92*	0.96	0.97	1.10 ⁺	1.02	0.99	1.03	0.99	0.94	0.98	0.90
Mother indigenous	0.87**	0.92*	0.98	0.90*	1.05	1.07	1.00	1.11*	0.97	1.00	0.92	0.78***
Both parents indigenous	1.20*	1.16*	1.00	1.05	0.81*	0.86 ⁺	0.98	0.88	1.03	1.12	0.97	1.68***
Control variables												
Study in a different municipality	0.29***	0.28***	0.79***	0.58***	1.29***	1.32***	1.10***	1.14***	1.38***	1.41***	1.11***	1.06**
Schools with additional tuition	0.83***	0.85***	0.80***	0.75***	1.63***	1.31***	1.40***	1.35***	0.87***	0.94*	0.85***	1.09*
Schools with selection process	0.47***	0.51***	0.48***	0.56***	1.79***	1.74***	2.28***	2.64***	2.68***	3.03***	3.27***	2.95***
School type												
Private-voucher	0.67***	0.64***	0.67***	1.22***	0.60***	0.74***	0.66***	0.41***	6.79***	6.98***	7.57***	10.60***
Private	0.32***	0.35***	0.31***	0.56***	0.91*	0.82***	1.05	0.55***	6.28***	6.57***	6.09***	12.68***
Urban school	0.48***	0.49***	0.57***	1.03	1.58***	1.55***	1.65***	2.00***	1.25***	1.19***	1.22***	0.58***
Numbers of schools per municipality	1.01	1.03 ⁺	0.98	0.99	1.00	1.02 ⁺	1.05**	0.96***	1.00	1.04***	1.05 ⁺	0.96**
Observations	97,634	140,492	142,192	109,124	97,133	139,984	141,587	108,841	96,299	138,521	140,274	108,935

***p < 0.001, **p < 0.01, *p < 0.05, ⁺p < 0.10.

choosing a school based on its proximity decreases as income increases in all grades, although the change is smaller in 8th and 10th grades. For instance, for a non-indigenous family in fourth grade the predicted probability decreases from about 0.4 in the lowest income level to about 0.3 in the highest income level. For grades 2 to 8 the difference in predicted probabilities between the lowest and the highest income level is statistically significant, because their 95% confidence intervals do not overlap. The figure also shows that the difference between families in

which both parents are indigenous and those in which neither parent is indigenous is very small. Overall in 10th grade proximity does not seem to be a relevant attribute for parents across incomes.

Fig. 9 shows that as income increases the average predicted probability of a family choosing a school based on its quality also increases. This change in predicted probability is bigger than in the case of proximity. For instance, in 2nd grade it rises from about 0.16 in the lowest quartile level to about 0.22 in the highest quartile, for non-

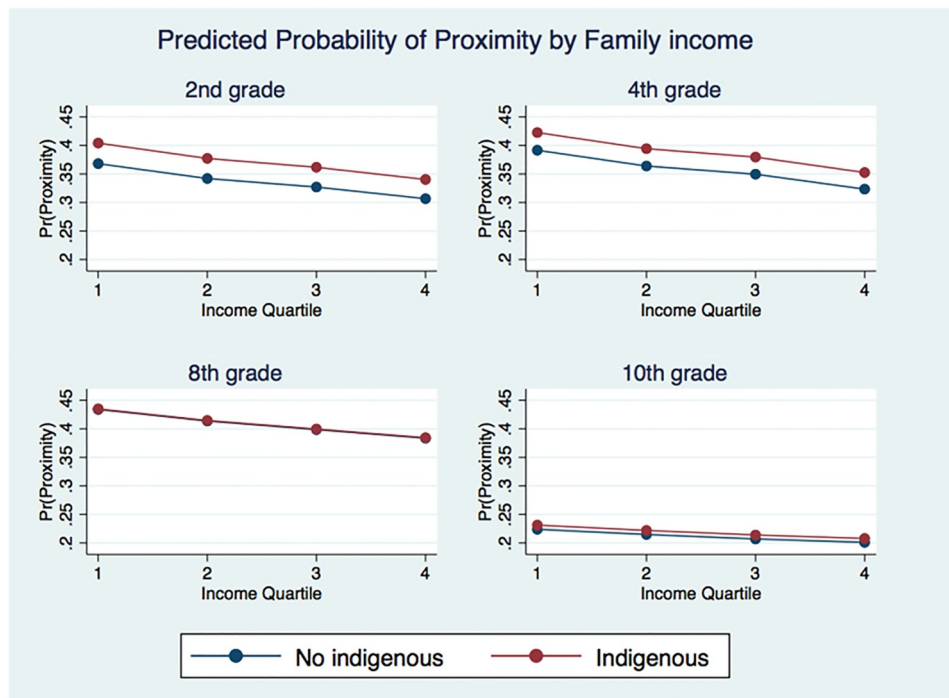


Fig. 8. Predicted probabilities for proximity attribute by income quartile and parents' ethnicity.

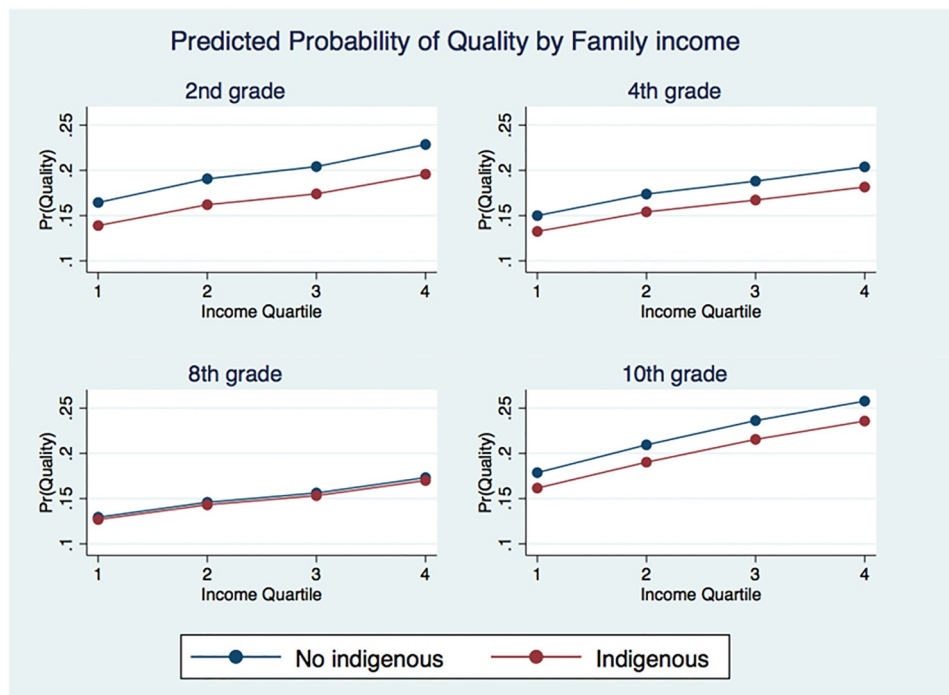


Fig. 9. Predicted probabilities for quality attribute by income quartile and parents' ethnicity.

indigenous families. In 10th grade it rises from about 0.17 to 0.26 in the same income range. The differences in predicted probabilities between the lowest and highest income level in all grades are statistically significant, because their 95% confidence intervals do not overlap. Fig. 9 also shows that non-indigenous families have a higher average predicted probability for quality than indigenous families for all income quartiles.

The second strategy we used to estimate predicted probabilities was to create six ideal types of pupils based on four socioeconomic

characteristics and estimate average predicted probabilities for students within each ideal case. The four characteristics used to create the ideal cases were: ethnicity, area of residence, parents' education and family income level.

Table 4 shows a summary of the 6 ideal cases and their characteristics. Ideal case 1 is the case with the lowest socioeconomic levels. It corresponds to a family in which both parents are indigenous, living in a rural area, the mother does not have a high school degree and the family's income is in the first quartile. Socioeconomic characteristics

Table 4
Ideal cases for the estimation of average predicted probabilities.

Ideal case	Both parents indigenous	Rural area	Mother has high school?	Income quartile
1	Yes	Yes	No	1
2	No	Yes	No	1
3	No	No	No	1
4	No	No	Yes	1
5	No	No	Yes	2 & 3
6	No	No	Yes	4

improve with each ideal case until ideal case 6, which is the case with the highest socioeconomic level. It corresponds to a family in which neither parent is indigenous, living in an urban setting, the mother has at least a high school degree, and the family's income is in the top quartile.

Fig. 10 presents average predicted probabilities for proximity for each ideal type by grade. The graph includes the point estimate along with confidence intervals. However, most of the confidence intervals are narrow, which makes them indistinguishable from the point estimate. In the case of 2nd grade the predicted probability of choosing a school based on its proximity goes from 0.6 for the first ideal type, i.e. an indigenous pupil, living in a rural area, whose mother did not finish high school and have low income (quartile 1), to around 0.2 for the sixth ideal type, i.e. a pupil whose parents are not indigenous, living in a urban area, with her mother having completed high school and a high income (quartile 4). 4th and 8th grade show similar changes. In 10th grade, as expected, the probabilities of prioritizing proximity are lower in low socioeconomic levels as compared to other grades, but they also decrease as socioeconomic status increases, as indicated by these ideal types. The differences between each pair for adjacent ideal types are all statistically significant, because their 95% confidence intervals do not overlap. These results confirm that, as socioeconomic status increases, families are less likely to choose a school based on its proximity.

Fig. 11 shows predicted probabilities for quality over ideal types by grade. It shows that the predicted probability of choosing a school based on its quality increases as socioeconomic status, as measured by

these ideal types, improves. For instance in second grade it increases from 0.06 in ideal type 1, which represents the lowest socioeconomic level, to 0.26 in the type with the highest socioeconomic level. This change is similar across all grades. Also, in this case the difference between low socioeconomic status indigenous and non-indigenous families is not statistically significant. These results provide further evidence for the assertion that as socioeconomic status increases, families are more likely to choose a school based on its quality.

Discussion and conclusions

Advocates of free school choice argue that individuals are the basic maximizing units, and therefore an educational system which provides options, like voucher school programs, would imply utility-maximizing actions by parents and pupils (Betts & Loveless, 2005). In theory, in a school choice program, parents would choose the best option for their children: schools with remarkable achievements, high test scores and teachers of excellence, among other aspects. However, if parents from different socioeconomic and demographic backgrounds do not consider similar aspects in choosing a school, due to their differing resources and available options, free school choice programs could lead to pupil separation based on their parents' possibilities, encouraging school segregation.

The evidence collected on parental preferences faces several limitations. First, in most countries, school choice programs are targeted for specific groups (low-income families, minority groups); furthermore, voucher school programs have been implemented for short periods of time or in very particular conditions. Therefore, there is not enough information to test how parents select schools in free school choice programs. To overcome this problem, we analyzed the case of the unrestricted, nationwide voucher school program implemented in Chile.

We analyzed parents' preferences in grades equivalent to elementary, middle and high school, and conclude that parents tend to consider three main aspects when they choose a school: proximity, quality and religious education. But these aspects are not similarly considered by different socioeconomic or ethnic groups. While proximity is more important among low income, indigenous and less educated parents, it

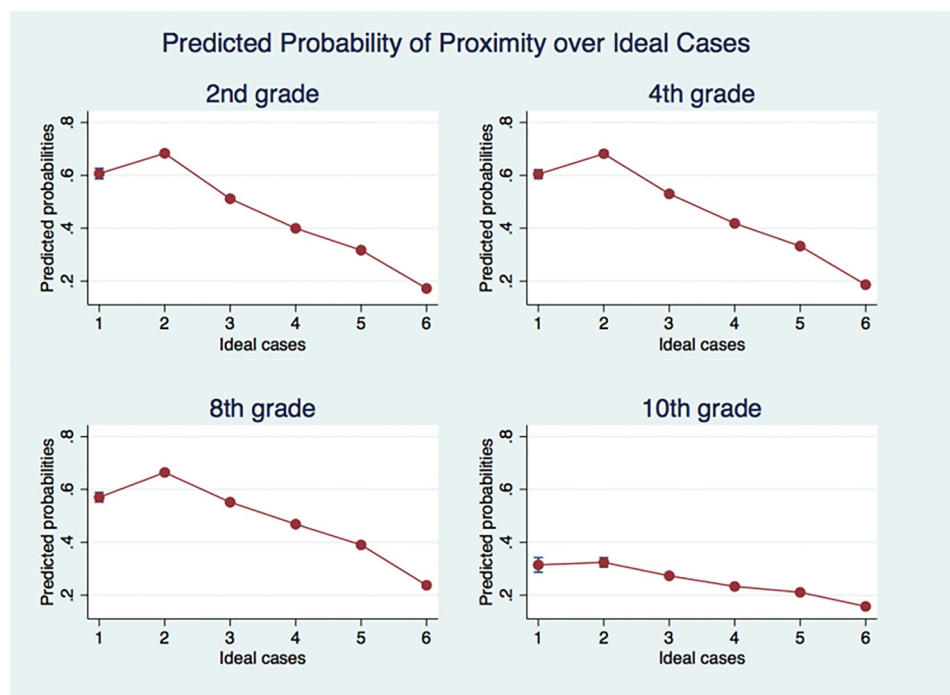


Fig. 10. Predicted probabilities for proximity by ideal cases.

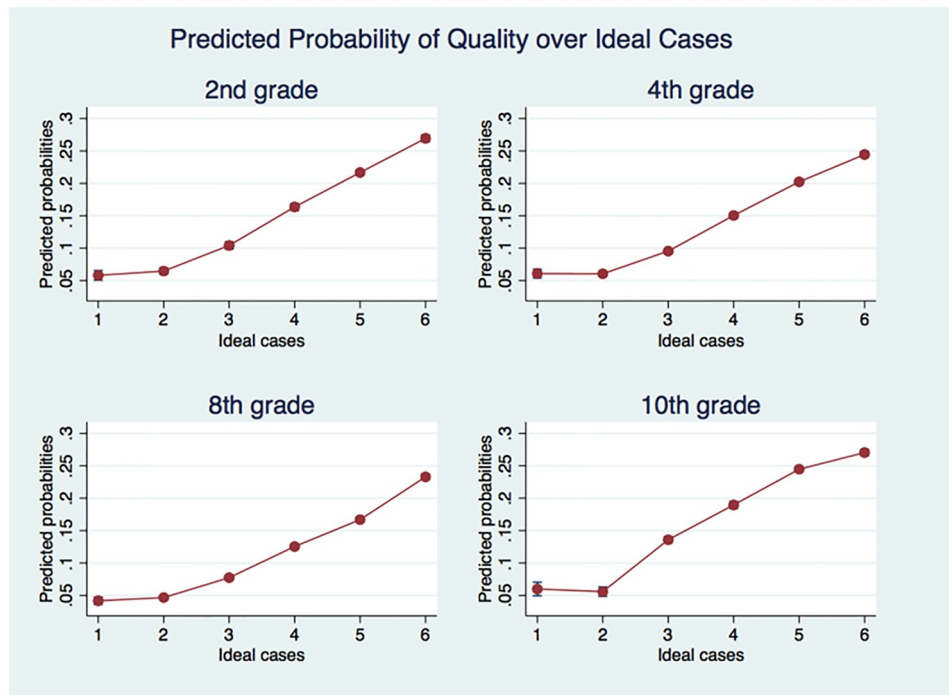


Fig. 11. Predicted probabilities for quality by ideal cases.

becomes less important as the family's income level increases. Secondly, as family income and educational level increase, school quality becomes more important in choice of school. Finally, as the mother's years of schooling increase, the school's religious education tends to acquire greater importance in selecting a school.

These results show that, even though in theory every parent and pupil would benefit by the free school choice program, at the national level in a nationwide program, medium and high-income families are the only ones that benefit, in the sense that those families are able to choose a school based on other factors, such as perceived school quality, because they do not seem to be limited to the choice of a nearby school. Moreover, this research reinforces the evidence collected in previous studies in specific Chilean localities, indicating that parents of a low socioeconomic level are at disadvantage when choosing a school due to their lower possibilities of paying for tuition fees and transportation costs, and of the child passing private-voucher schools' admission processes (Román & Corvalán, 2016; Córdoba, 2014; Flores & Carrasco, 2013). In a widely segregated educational system such as the Chilean system, this disadvantage restricts these families to choosing nearby schools, which are usually of low quality. The new regulation that bans co-payment from private-voucher schools (Inclusion Law) will have residual effects unless the supply of high-quality subsidized schools increases in vulnerable areas, becoming accessible for low income families.

This study is somewhat limited by the data. First, parents' reports of their preferences may be affected by temporality. We do not know for how long pupils have attended the school for which they are reported, and therefore we do not know if the reason why parents selected the school is their original one or has changed over time. Second, parents' answers may be affected by the methodology. Previous studies have identified that parents report different reasons why they chose a school if they answer a questionnaire (like in the data we use) or if they are interviewed face to face. Finally, parents' answers may be affected by social desirability. Previous qualitative studies have shown that parents consider school composition as an important factor in deciding which school to choose. However, this was only reported by 4 percent in our sample (Table 2).

Despite these limitations, this study reveals the importance of carefully analysing parents' preferences, and how those differ between families of different socioeconomic status, in order to judge the merits of free school choice programs and the alternatives.

Credit authorship contribution statement

Alvaro Hofflinger: Conceptualization, Methodology, Data curation, Writing - original draft. **Denisse Gelber:** Conceptualization, Writing - review & editing. **Santiago Tellez Cañas:** Methodology, Data curation, Writing - review & editing.

CRediT authorship contribution statement

Alvaro Hofflinger: Conceptualization, Methodology, Data curation, Writing - original draft. **Denisse Gelber:** Conceptualization, Writing - review & editing. **Santiago Tellez Cañas:** Methodology, Data curation, Writing - review & editing.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.econedurev.2019.101946](https://doi.org/10.1016/j.econedurev.2019.101946).

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